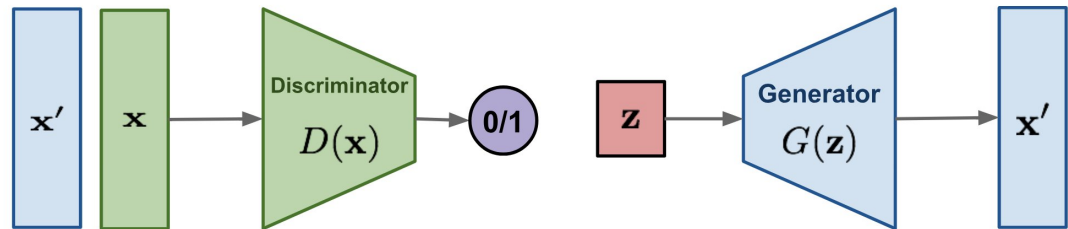


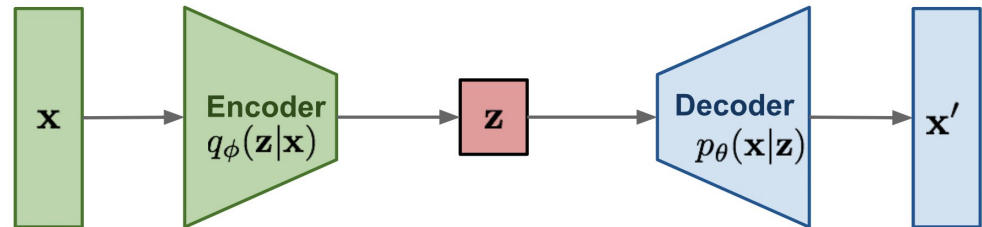
Введение в диффузионные модели



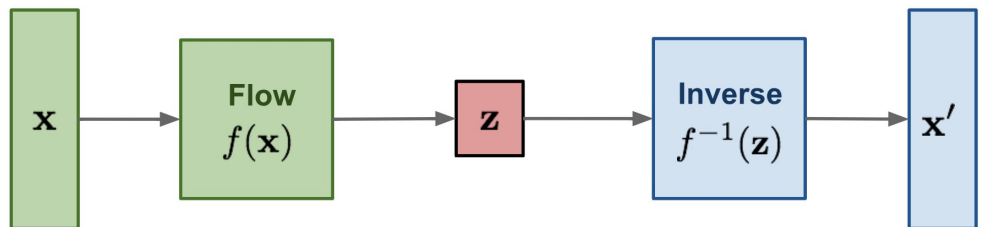
GAN: minimax the classification error loss.



VAE: maximize ELBO.

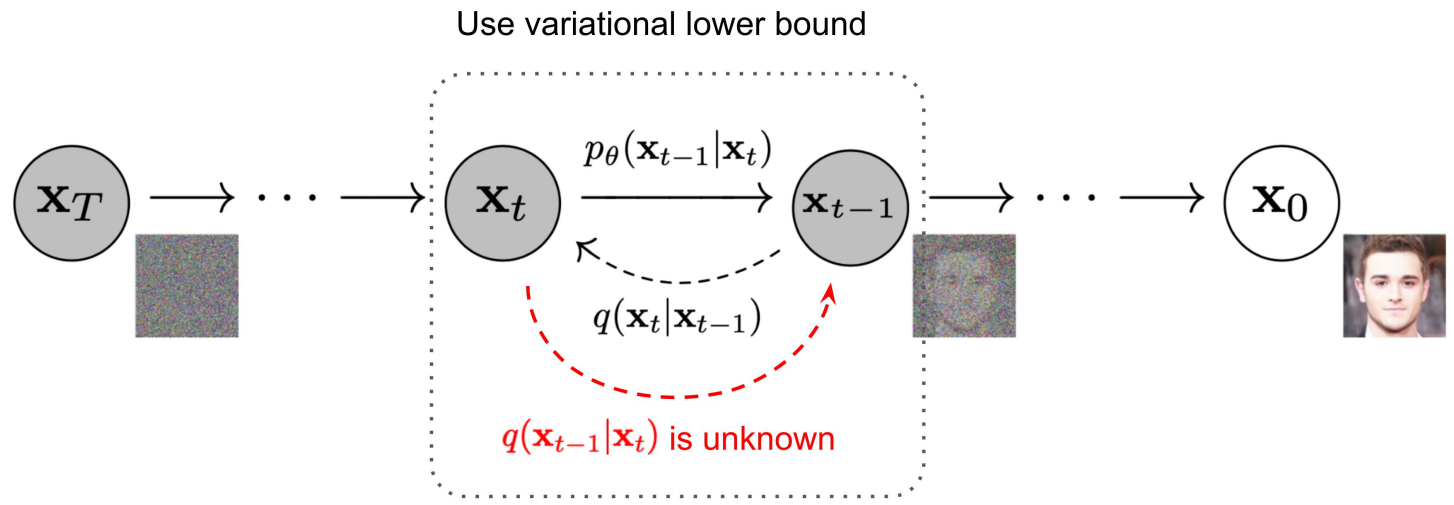


Flow-based generative models: minimize the negative log-likelihood



Модель	Идея	Достоинства	Недостатки
GAN	Генеративно-состозательный подход на основе двух соревнующихся нейросетей	Высокое качество изображений	Mode collapsing Сложность в обучении Избыточность
VAE	Байесовская модель сжатия и декомпрессии изображения	Лёгкость в обучении Вероятностная модель	Размытость картинок Избыточность
NF	Обратимое преобразование между данными и табличным распределением	Лёгкость в обучении Доступ к плотности распределения (Exlicit Probabilistic Model)	Ограниченность из-за требований к обратимости и якобиану Точечная оценка





Denoising diffusion models

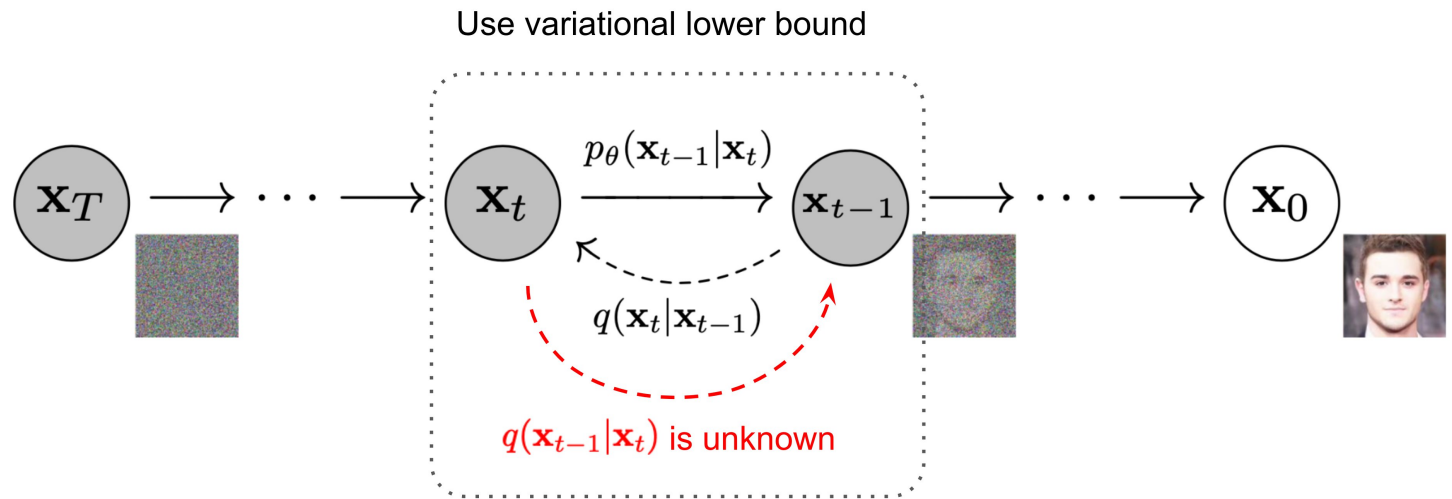
- **Forward / noising process**

- Sample data $p(\mathbf{x}_0) \rightarrow$ turn to noise



- **Reverse / denoising process**

- Sample noise $p_T(\mathbf{x}_T) \rightarrow$ turn into data



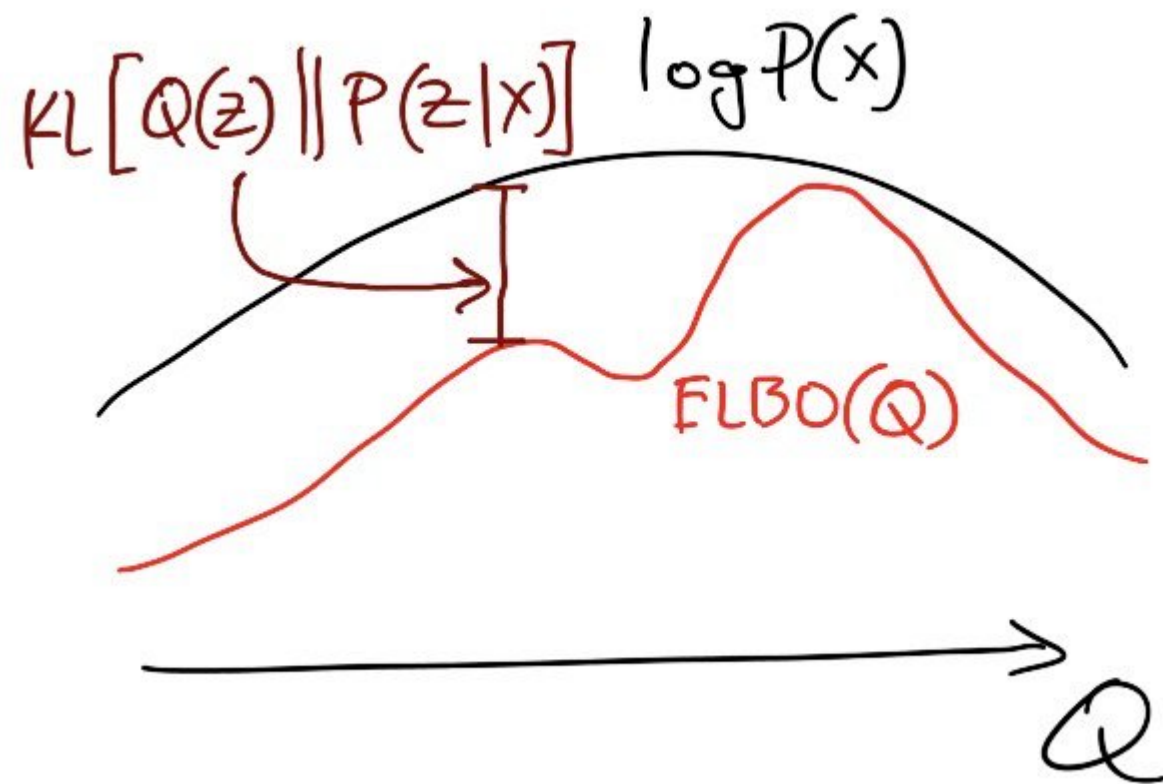
Forward process

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}) \quad q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1})$$

Backward process?

Диффузионные модели. Теория





Variational Lower Bound (VLB)

$$\begin{aligned} L_{\text{CE}} &= -\mathbb{E}_{q(\mathbf{x}_0)} \log p_{\theta}(\mathbf{x}_0) \\ &= -\mathbb{E}_{q(\mathbf{x}_0)} \log \left(\int p_{\theta}(\mathbf{x}_{0:T}) d\mathbf{x}_{1:T} \right) \\ &= -\mathbb{E}_{q(\mathbf{x}_0)} \log \left(\int q(\mathbf{x}_{1:T}|\mathbf{x}_0) \frac{p_{\theta}(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} d\mathbf{x}_{1:T} \right) \\ &= -\mathbb{E}_{q(\mathbf{x}_0)} \log \left(\mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \frac{p_{\theta}(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right) \\ &\leq -\mathbb{E}_{q(\mathbf{x}_{0:T})} \log \frac{p_{\theta}(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \\ &= \mathbb{E}_{q(\mathbf{x}_{0:T})} \left[\log \frac{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}{p_{\theta}(\mathbf{x}_{0:T})} \right] = L_{\text{VLB}} \end{aligned}$$



Variational Lower Bound (VLB)

$$\begin{aligned}
 L_{\text{VLB}} &= \mathbb{E}_{q(\mathbf{x}_{0:T})} \left[\log \frac{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T})} \right] \\
 &= \mathbb{E}_q \left[\log \frac{\prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1})}{p_\theta(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} \right] \\
 &= \mathbb{E}_q \left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=1}^T \log \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} \right] \\
 &= \mathbb{E}_q \left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^T \log \frac{q(\mathbf{x}_t|\mathbf{x}_{t-1})}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_\theta(\mathbf{x}_0|\mathbf{x}_1)} \right] \\
 &= \mathbb{E}_q \left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^T \log \left(\frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} \cdot \frac{q(\mathbf{x}_t|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_0)} \right) + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_\theta(\mathbf{x}_0|\mathbf{x}_1)} \right] \\
 &= \mathbb{E}_q \left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^T \log \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} + \sum_{t=2}^T \log \frac{q(\mathbf{x}_t|\mathbf{x}_0)}{q(\mathbf{x}_{t-1}|\mathbf{x}_0)} + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_\theta(\mathbf{x}_0|\mathbf{x}_1)} \right] \\
 &= \mathbb{E}_q \left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^T \log \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} + \log \frac{q(\mathbf{x}_T|\mathbf{x}_0)}{q(\mathbf{x}_1|\mathbf{x}_0)} + \log \frac{q(\mathbf{x}_1|\mathbf{x}_0)}{p_\theta(\mathbf{x}_0|\mathbf{x}_1)} \right] \\
 &= \mathbb{E}_q \left[\log \frac{q(\mathbf{x}_T|\mathbf{x}_0)}{p_\theta(\mathbf{x}_T)} + \sum_{t=2}^T \log \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)} - \log p_\theta(\mathbf{x}_0|\mathbf{x}_1) \right] \\
 &= \mathbb{E}_q \left[\underbrace{D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p_\theta(\mathbf{x}_T))}_{L_T} + \sum_{t=2}^T \underbrace{D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{L_{t-1}} - \log p_\theta(\mathbf{x}_0|\mathbf{x}_1) \right]
 \end{aligned}$$



$$\begin{aligned}
q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) &= q(\mathbf{x}_t|\mathbf{x}_{t-1}, \mathbf{x}_0) \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)} \\
&\propto \exp \left(-\frac{1}{2} \left(\frac{(\mathbf{x}_t - \sqrt{\alpha_t} \mathbf{x}_{t-1})^2}{\beta_t} + \frac{(\mathbf{x}_{t-1} - \sqrt{\bar{\alpha}_{t-1}} \mathbf{x}_0)^2}{1 - \bar{\alpha}_{t-1}} - \frac{(\mathbf{x}_t - \sqrt{\bar{\alpha}_t} \mathbf{x}_0)^2}{1 - \bar{\alpha}_t} \right) \right) \\
&= \exp \left(-\frac{1}{2} \left(\frac{\mathbf{x}_t^2 - 2\sqrt{\alpha_t} \mathbf{x}_t \mathbf{x}_{t-1} + \alpha_t \mathbf{x}_{t-1}^2}{\beta_t} + \frac{\mathbf{x}_{t-1}^2 - 2\sqrt{\bar{\alpha}_{t-1}} \mathbf{x}_0 \mathbf{x}_{t-1} + \bar{\alpha}_{t-1} \mathbf{x}_0^2}{1 - \bar{\alpha}_{t-1}} - \frac{(\mathbf{x}_t - \sqrt{\bar{\alpha}_t} \mathbf{x}_0)^2}{1 - \bar{\alpha}_t} \right) \right) \\
&= \exp \left(-\frac{1}{2} \left(\left(\frac{\alpha_t}{\beta_t} + \frac{1}{1 - \bar{\alpha}_{t-1}} \right) \mathbf{x}_{t-1}^2 - \left(\frac{2\sqrt{\alpha_t}}{\beta_t} \mathbf{x}_t + \frac{2\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_{t-1}} \mathbf{x}_0 \right) \mathbf{x}_{t-1} + C(\mathbf{x}_t, \mathbf{x}_0) \right) \right)
\end{aligned}$$



Forward process. Reparametrization trick

$$\begin{aligned}\mathbf{x}_t &= \sqrt{\alpha_t} \mathbf{x}_{t-1} + \sqrt{1 - \alpha_t} \boldsymbol{\epsilon}_{t-1} && \text{;where } \boldsymbol{\epsilon}_{t-1}, \boldsymbol{\epsilon}_{t-2}, \dots \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \\ &= \sqrt{\alpha_t \alpha_{t-1}} \mathbf{x}_{t-2} + \sqrt{1 - \alpha_t \alpha_{t-1}} \bar{\boldsymbol{\epsilon}}_{t-2} && \text{;where } \bar{\boldsymbol{\epsilon}}_{t-2} \text{ merges two Gaussians (*).} \\ &= \dots \\ &= \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon} \\ q(\mathbf{x}_t | \mathbf{x}_0) &= \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})\end{aligned}$$

$$\alpha_t = 1 - \beta_t \quad \bar{\alpha}_t = \prod_{i=1}^t \alpha_i$$



From the previous slide,

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$$

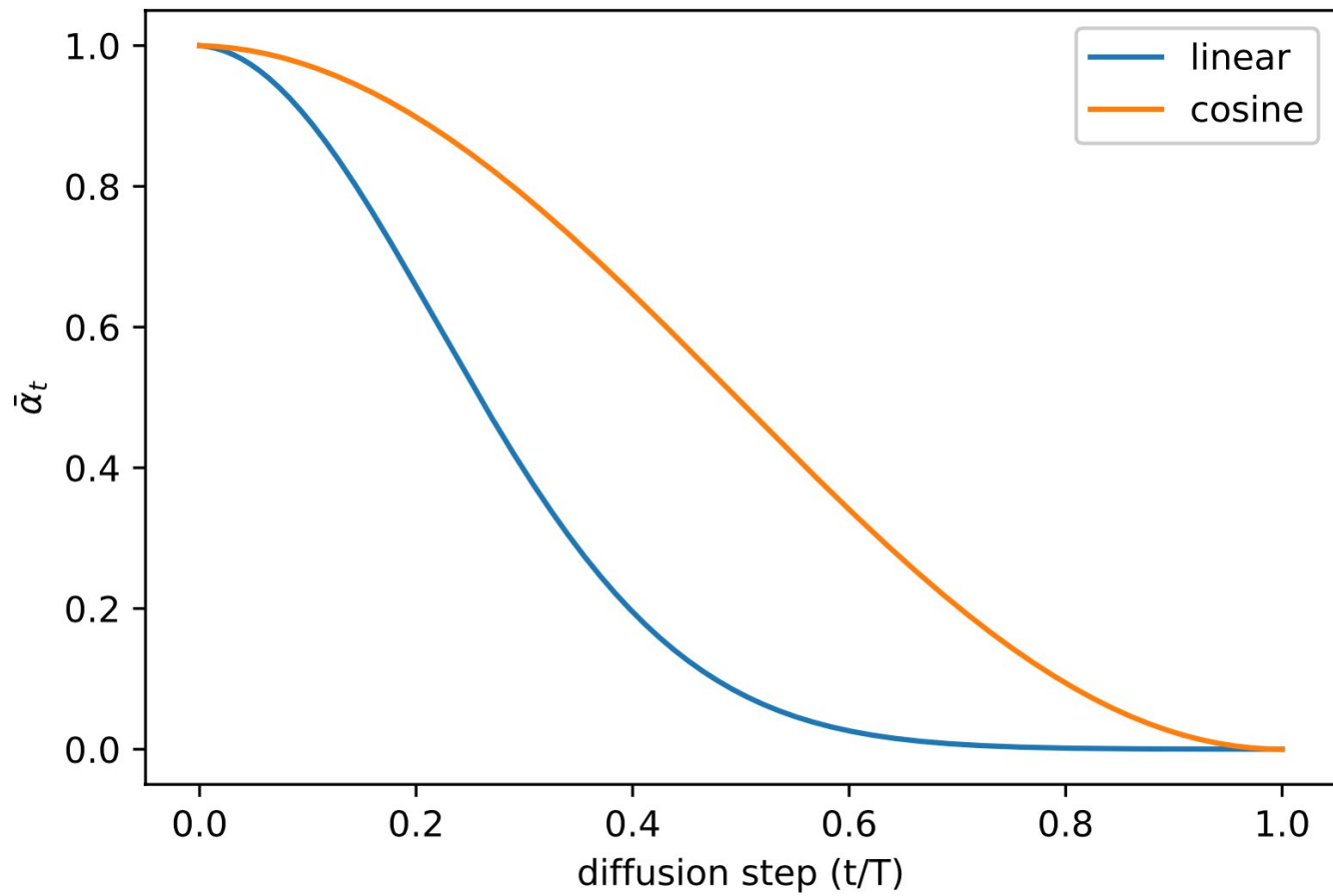
$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \epsilon_t (1 - \bar{\alpha}_t)$$
$$x_0 = \frac{x_t - \epsilon_t (1 - \bar{\alpha}_t)}{\sqrt{\bar{\alpha}_t}}$$

Then

$$\begin{aligned} \bar{\beta}_t &= 1 / \left(\frac{\alpha_t}{\beta_t} + \frac{1}{1 - \bar{\alpha}_{t-1}} \right) = 1 / \left(\frac{\alpha_t - \bar{\alpha}_t + \beta_t}{\beta_t (1 - \bar{\alpha}_{t-1})} \right) = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \cdot \beta_t \\ \tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) &= \left(\frac{\sqrt{\alpha_t}}{\beta_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_{t-1}} \mathbf{x}_0 \right) / \left(\frac{\alpha_t}{\beta_t} + \frac{1}{1 - \bar{\alpha}_{t-1}} \right) \\ &= \left(\frac{\sqrt{\alpha_t}}{\beta_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_{t-1}} \mathbf{x}_0 \right) \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \cdot \beta_t \\ &= \frac{\sqrt{\alpha_t} (1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}} \beta_t}{1 - \bar{\alpha}_t} \mathbf{x}_0 = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_t \right) \end{aligned}$$

$$q(x_{t-1} | x_t, x_0) = \mathcal{N}(x_{t-1} | \tilde{\mu}_t(x_t, x_0), \tilde{\beta}_t)$$





$$L_t = D_{\text{KL}}(q(\mathbf{x}_t | \mathbf{x}_{t+1}, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_t | \mathbf{x}_{t+1})) \text{ for } 1 \leq t \leq T - 1$$

Let $p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{x}_t, t), \boldsymbol{\Sigma}_\theta(\mathbf{x}_t, t))$

$$\boldsymbol{\Sigma}_\theta(\mathbf{x}_t, t) = \tilde{\beta}_t$$

$$\boldsymbol{\mu}_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right)$$

$$\begin{aligned} L_t &= \mathbb{E}_{\mathbf{x}_0, \boldsymbol{\epsilon}} \left[\frac{1}{2 \|\boldsymbol{\Sigma}_\theta(\mathbf{x}_t, t)\|_2^2} \|\tilde{\boldsymbol{\mu}}_t(\mathbf{x}_t, \mathbf{x}_0) - \boldsymbol{\mu}_\theta(\mathbf{x}_t, t)\|^2 \right] \\ &= \mathbb{E}_{\mathbf{x}_0, \boldsymbol{\epsilon}} \left[\frac{1}{2 \|\boldsymbol{\Sigma}_\theta\|_2^2} \left\| \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \boldsymbol{\epsilon}_t \right) - \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right) \right\|^2 \right] \\ &= \mathbb{E}_{\mathbf{x}_0, \boldsymbol{\epsilon}} \left[\frac{(1 - \alpha_t)^2}{2 \alpha_t (1 - \bar{\alpha}_t) \|\boldsymbol{\Sigma}_\theta\|_2^2} \|\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\|^2 \right] \\ &= \mathbb{E}_{\mathbf{x}_0, \boldsymbol{\epsilon}} \left[\frac{(1 - \alpha_t)^2}{2 \alpha_t (1 - \bar{\alpha}_t) \|\boldsymbol{\Sigma}_\theta\|_2^2} \|\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}_t, t)\|^2 \right] \end{aligned}$$

$$\begin{aligned} L_t^{\text{simple}} &= \mathbb{E}_{t \sim [1, T], \mathbf{x}_0, \boldsymbol{\epsilon}_t} \left[\|\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\|^2 \right] \\ &= \mathbb{E}_{t \sim [1, T], \mathbf{x}_0, \boldsymbol{\epsilon}_t} \left[\|\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}_t, t)\|^2 \right] \end{aligned}$$



Algorithm 1 Training

```
1: repeat  
2:    $\mathbf{x}_0 \sim q(\mathbf{x}_0)$   
3:    $t \sim \text{Uniform}(\{1, \dots, T\})$   
4:    $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$   
5:   Take gradient descent step on  
        $\nabla_{\theta} \|\epsilon - \epsilon_{\theta}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t)\|^2$   
6: until converged
```

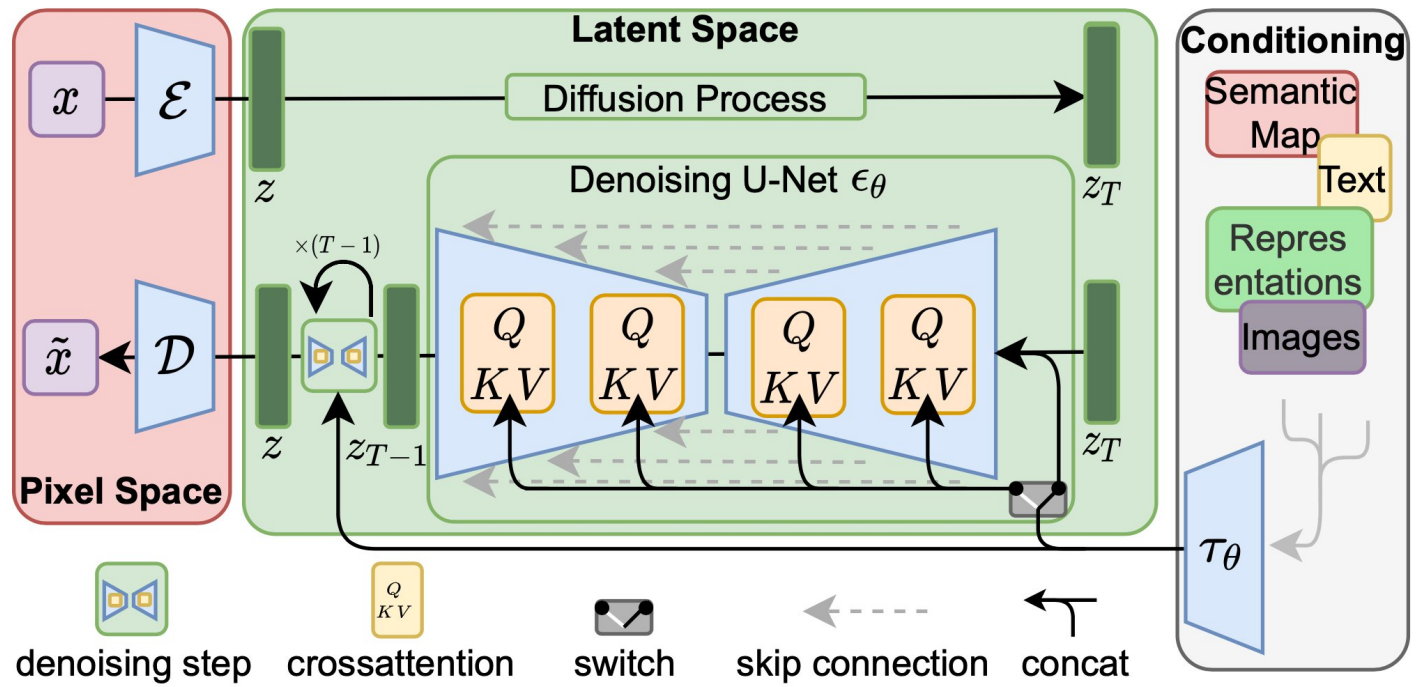
Algorithm 2 Sampling

```
1:  $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$   
2: for  $t = T, \dots, 1$  do  
3:    $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  if  $t > 1$ , else  $\mathbf{z} = \mathbf{0}$   
4:    $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_{\theta}(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$   
5: end for  
6: return  $\mathbf{x}_0$ 
```



Диффузионные модели. Advanced tricks





$$\begin{aligned}
\mathbf{x}_{t-1} &= \sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_{t-1}}\boldsymbol{\epsilon}_{t-1} \\
&= \sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2}\boldsymbol{\epsilon}_t + \sigma_t\boldsymbol{\epsilon} \\
&= \sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0}{\sqrt{1 - \bar{\alpha}_t}} + \sigma_t\boldsymbol{\epsilon} \\
q_\sigma(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) &= \mathcal{N}(\mathbf{x}_{t-1}; \sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0}{\sqrt{1 - \bar{\alpha}_t}}, \sigma_t^2\mathbf{I})
\end{aligned}$$

Recall that in $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \bar{\boldsymbol{\mu}}(\mathbf{x}_t, \mathbf{x}_0), \bar{\beta}_t\mathbf{I})$, therefore we have:

$$\bar{\beta}_t = \sigma_t^2 = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \cdot \beta_t$$

Let

$$\sigma_t^2 = \eta \cdot \tilde{\beta}_t$$

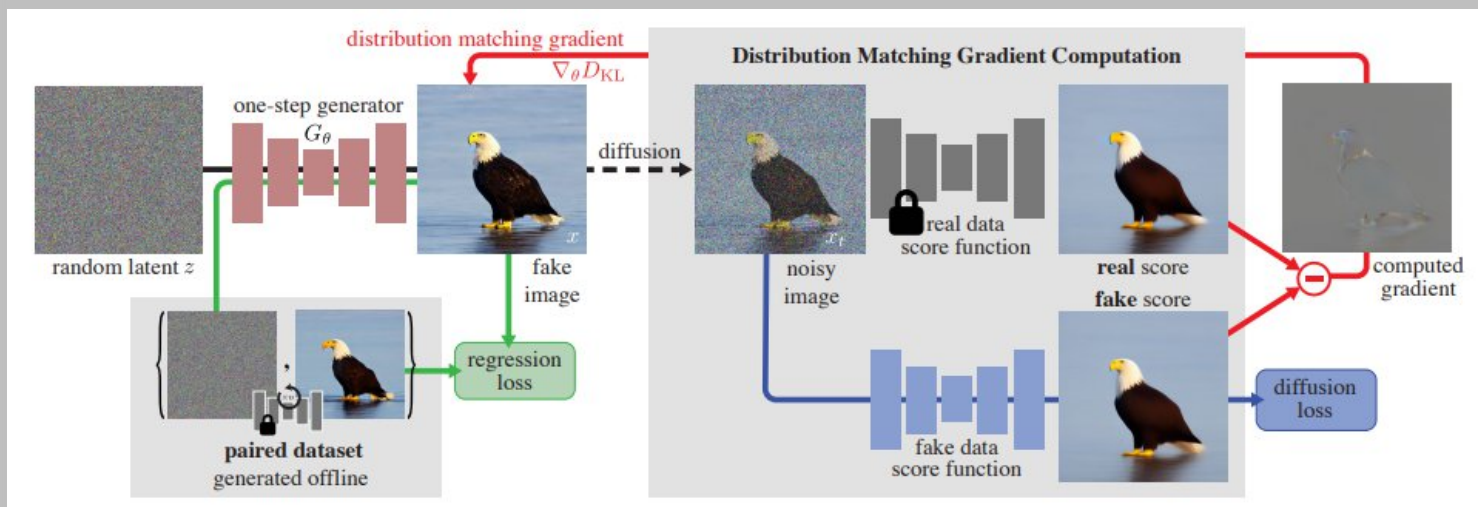
Then for $\eta = 0$

$$q_{\sigma, \tau}(\mathbf{x}_{\tau_{i-1}}|\mathbf{x}_{\tau_i}, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{\tau_{i-1}}; \sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \frac{\mathbf{x}_{\tau_i} - \sqrt{\bar{\alpha}_t}\mathbf{x}_0}{\sqrt{1 - \bar{\alpha}_t}}, \sigma_t^2\mathbf{I})$$

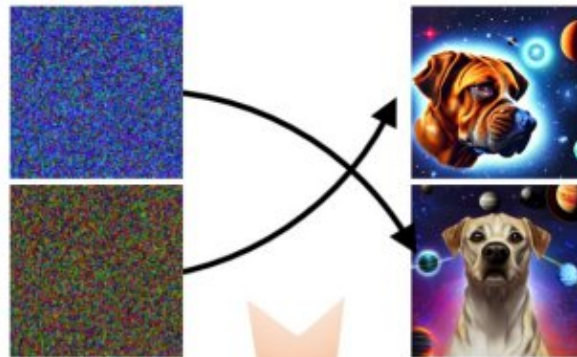


S	CIFAR10 (32×32)					CelebA (64×64)					
	10	20	50	100	1000	10	20	50	100	1000	
η	0.0	13.36	6.84	4.67	4.16	4.04	17.33	13.73	9.17	6.53	3.51
	0.2	14.04	7.11	4.77	4.25	4.09	17.66	14.11	9.51	6.79	3.64
	0.5	16.66	8.35	5.25	4.46	4.29	19.86	16.06	11.01	8.09	4.28
	1.0	41.07	18.36	8.01	5.78	4.73	33.12	26.03	18.48	13.93	5.98
$\hat{\sigma}$	367.43	133.37	32.72	9.99	3.17	299.71	183.83	71.71	45.20	3.26	



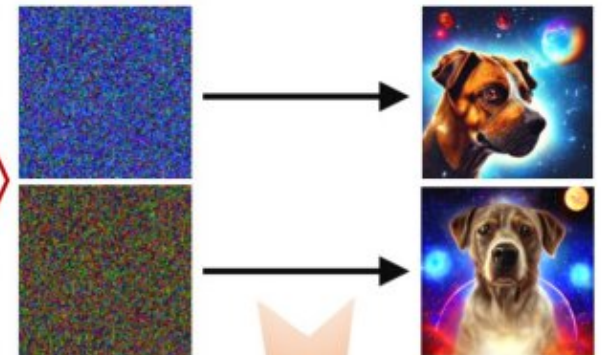


Original Model
Curved Trajectories; **Bad** Coupling

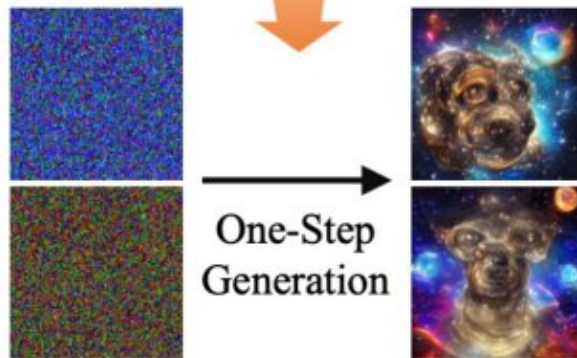


Reflow

Reflowed Model
Straighter Trajectories; **Good** Coupling



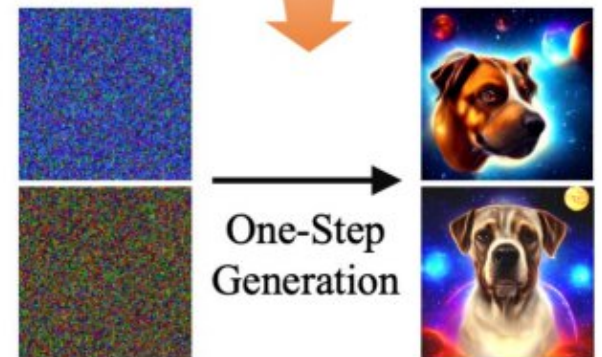
Distill



One-Step
Generation

 **Failure:** blurry, low-quality

Distill



One-Step
Generation

 **Success:** clear, high-quality



Classifier guidance

$$\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t) = \mathbb{E}_{q(\mathbf{x}_0)} [\nabla_{\mathbf{x}_t} q(\mathbf{x}_t | \mathbf{x}_0)] = \mathbb{E}_{q(\mathbf{x}_0)} \left[-\frac{\boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t)}{\sqrt{1 - \bar{\alpha}_t}} \right] = -\frac{\boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t)}{\sqrt{1 - \bar{\alpha}_t}}$$

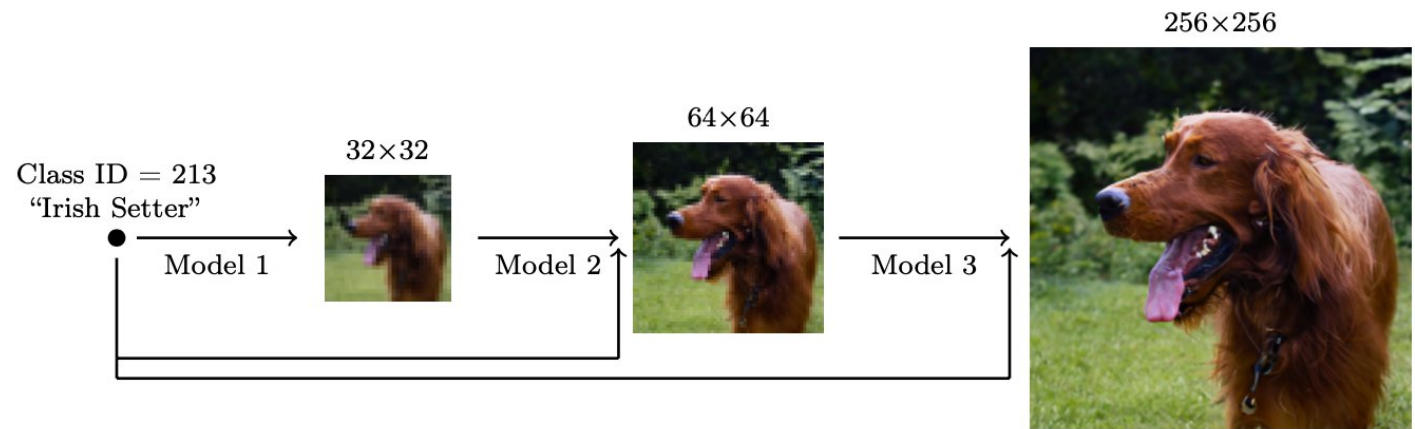
$$\begin{aligned} \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t, y) &= \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log q(y | \mathbf{x}_t) \\ &\approx -\frac{1}{\sqrt{1 - \bar{\alpha}_t}} \boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t) + \nabla_{\mathbf{x}_t} \log f_{\phi}(y | \mathbf{x}_t) \\ &= -\frac{1}{\sqrt{1 - \bar{\alpha}_t}} (\boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t) - \sqrt{1 - \bar{\alpha}_t} \nabla_{\mathbf{x}_t} \log f_{\phi}(y | \mathbf{x}_t)) \end{aligned}$$

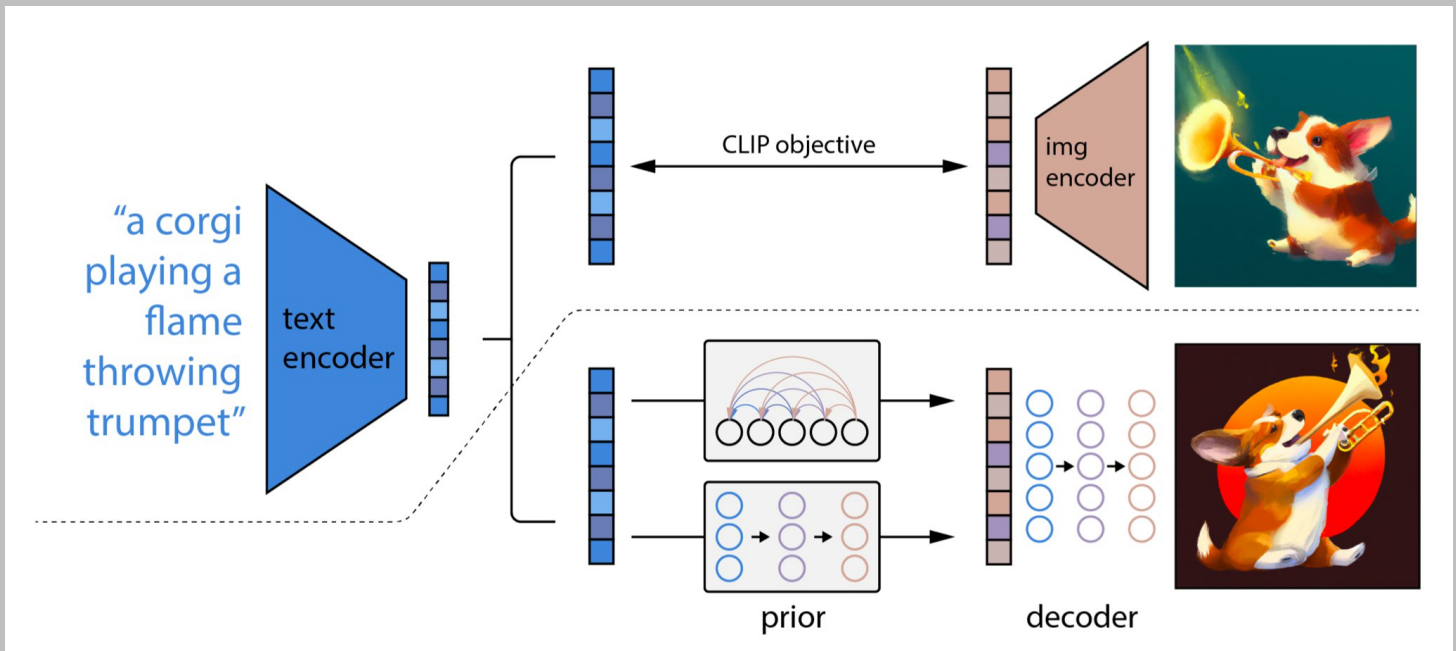


Classifier-free guidance

$$\begin{aligned}\nabla_{\mathbf{x}_t} \log p(y|\mathbf{x}_t) &= \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t|y) - \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t) \\ &= -\frac{1}{\sqrt{1-\bar{\alpha}_t}} \left(\boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t, y) - \boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t) \right) \\ \bar{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t, y) &= \boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t, y) - \sqrt{1-\bar{\alpha}_t} w \nabla_{\mathbf{x}_t} \log p(y|\mathbf{x}_t) \\ &= \boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t, y) + w \left(\boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t, y) - \boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t) \right) \\ &= (w+1) \boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t, y) - w \boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t)\end{aligned}$$







Диффузионные модели. Приложения



Image2Image. Текстовая инверсия



Inpainting



Outpainting



Outpainting



- **Text to video**
- **Text to 3D**
- **Text to motion**
- **Image Search & Reverse Image Search**
- **Zoom In/Out**

